

# Query Intent Resolver

V1.1 | Technical closeout and integration handoff

## Delivery decision

The query-only engineering package is complete. Internal quality gates are not all met. Live MascotGO deployment is not authorized: concrete handlers, product approval and independently labeled real traffic remain external requirements.

## What was finished

The submitted 55-row review was verified against the quarantine queue. 15 labels were retained unchanged; 30 were separately policy-adjudicated; 10 context-dependent queries were excluded. The original reviewer CSV and every original V1 file are unchanged. The strict new fit contains 7,071 queries.

## Model selection and evidence

The selected candidate is linearsvc. Three fixed candidates were evaluated using grouped training out-of-fold predictions. Model, routing threshold and optional lookup rule were fixed before reading evaluation labels. The legacy 300-query benchmark is historical regression evidence, not a new blind test. V1.1 has zero exact and zero heuristic template-family overlaps with its fit; semantic overlap beyond this heuristic remains possible.

| Measure                 | Historical 300 | Core applicability 100 |
|-------------------------|----------------|------------------------|
| Accuracy                | 80.33%         | 62.00%                 |
| Macro F1                | 0.8041         | 0.6175                 |
| False SC / predicted SC | 14.52%         | 20.00%                 |
| Short-circuit recall    | 70.67%         | 64.00%                 |

## **Applicability and runtime**

The balanced 100-case assistant-authored core suite achieved 62.00% accuracy and 0.6175 macro F1. Twenty boundary cases are reported separately. These cases are neither independent human testing nor real-user traffic. The runtime passed 134/134 contract, input-validation, batch, parity and threading checks.

## **Confidence and cost**

Learned confidence is the probability of the emitted route, not the probability of a discarded prediction. Guard fallback confidence is zero to indicate unknown. No external model API calls or LLM tokens were used in this closeout; infrastructure, CPU and downstream costs are not claimed to be zero.

## **Integration boundary**

The API accepts raw query\_text and returns exactly route and confidence. Persona, page, filters, context and session do not influence V1.1 predictions. Python, CLI, FastAPI, OpenAPI, Docker and explicit four-handler binding examples are supplied. There are no fabricated endpoint URLs, credentials or claimed live Foundry deployments.

## **Peter's remaining decisions**

Confirm the invocation surface, route-to-handler mapping, indexed-field coverage, optional request fields, preferred deployment environment and an authorized real-query or shadow sample. Approve error tolerances only after independent validation. Issue #5 remains the external product decision record.

## **Use this package**

Start with v1\_1/docs/REVIEWER\_BRIEF.html and TECHNICAL\_CLOSEOUT.md. Exact metrics, row-level predictions, failures, review dispositions, hashes and environment are under v1\_1/artifacts/. Reproduce with PYTHONPATH=v1\_1 python -m qir\_release.build --repo-root . using the pinned requirements. The previous V1.0.1 remains available for rollback and comparison.